Economic Disadvantage and Mass Shootings Across U.S. States, 2018 to 2024

A state-level panel analysis

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Under review, *Injury Epidemiology*

Mass shootings roughly doubled after 2019 and stayed high

Year	2018	2019	2020	2021	2022	2023	2024
Mass shootings, 50 states	329	406	602	673	633	649	491

2.0 times

as many in 2021 as in 2018

1.5 times

as many in 2024 as in 2018

With state characteristics held equal, each year from 2020 to 2023 had about 1.89 to 2.01 times the 2018 rate.

The surge was the largest signal in the data. This study documents it but cannot explain it.

The question

Public debate about prevention tends to focus on two levers: gun laws and mental illness. Much less attention goes to the economic conditions of the places where mass shootings happen.

I asked which state characteristics go with higher mass-shooting rates:

1. **Gun-law strength**
2. **Serious mental illness**
3. **Poverty**, as a sign of economic disadvantage

Knowing where mass shootings happen tells us where prevention can do the most good.

Data and method, in plain words

What I counted

Gun Violence Archive mass shootings: four or more people shot, not counting the shooter. All 50 states, 2018 to 2024: 3,783 incidents.

Only 5% had four or more people killed. In 46%, no victim died.

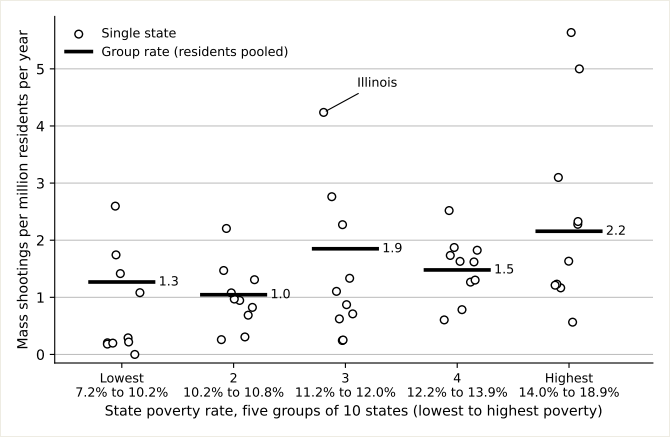
How I compared states

Rates per million residents. Poverty, gun-law strength and serious mental illness in one model, the year held equal, each from its official source.

Checks: leave out one state at a time, compare within regions, add income, add the share of Black residents.

This broad definition mostly captures community gun violence, not mass killings.

Higher-poverty states had more mass shootings



2.2
high poverty

1.3
low poverty

Mass shootings per million residents per year, 10 highest and 10 lowest poverty states: 1.7 times the rate. The high-poverty states held 16% of residents but 22% of incidents.

The rise was not steady: the middle group, with Illinois, was higher than the next group.

Observed rates, not adjusted; states, not people.

The poverty link held without any one state and within regions

With gun-law strength, mental illness and the year held equal, each percentage point of poverty went with about **15% more mass shootings per million residents** (95% CI 6% to 26% more).

- **Without any one state,** including Illinois, the state with the most incidents, the link stayed.
- **Within regions,** the estimate was nearly the same.
- **Before and after the surge:** 14% more per point in 2018 to 2019, 16% more in 2020 to 2024.

Of the three, only poverty went with more shootings; it held up best but not apart from low income.

But poverty travels with low income and segregation

- **Low income.** With income added, poverty went with 8% more per point (95% CI 6% fewer to 24% more), no longer clear. The data cannot tell the two apart.
- **Share of Black residents.** Each 10 percentage points more went with 57% more mass shootings. With it added, the poverty link was about half as large, 8% more per point, but still present. Race does not cause violence: at the state level this share marks segregation and concentrated disadvantage.
- **Other methods disagreed** on which economic measure matters most.
- An unmeasured factor as strong as the share of Black residents, the strongest factor I measured, could erase the poverty link.

I read poverty as one marker of economic disadvantage, not a single lever.

What did not show a clear or reliable link

Gun-law strength

4% more mass shootings per full letter grade (95% CI 10% fewer to 20% more): no clear link.

Within regions, stronger-law states had 19% more, but this rested on a few states. Without Illinois: 14% more (1% fewer to 31% more), no longer clear.

Serious mental illness

States with more mental illness did not have more mass shootings. They had 18% fewer per percentage point, but this was fragile and disappeared when the share of Black residents was added.

This is about states, not about any person.

No clear link is an absence of evidence, not evidence that gun laws do not work.

Mass shootings cluster in a few places

32%

of all incidents happened in just 20 cities or counties

74%

of the 375 incidents in Illinois happened in Chicago

Incidents occurred in 1,114 cities or counties, but the 20 places with the most held 1,217 of the 3,783.

One statewide grade or rate cannot capture the conditions, laws, enforcement and gun markets in these few places.

The state is a coarse unit for a problem this concentrated.

What this study cannot show

- **Associations, not causes.** It compares states, not people, and says nothing about individuals.
- **The state is a coarse unit,** and 50 states are a small number of comparisons.
- **The definition matters.** Only 5% of incidents had four or more killed. Results may differ for fatal or public mass shootings.
- **One snapshot of each state,** mostly from 2022 or 2023. It cannot explain the surge or show whether changes in poverty or laws came first.
- **Poverty, low income and segregation overlap,** and the study cannot say which matters most.

These results do not show that reducing poverty would reduce mass shootings.

What this means for policy

- **Where to focus.** Prevention and violence-intervention resources could go first to high-poverty states, to the cities and counties where shootings cluster, and to the people at highest risk there.
- **Economic conditions belong in the conversation,** alongside gun policy, not instead of it. Whether economic investment lowers mass shootings is worth testing in those places.
- **Gun laws need local data.** State data cannot settle whether they work. That needs city or county data linked to the laws in force at the time.
- **Mental-health care matters for its own reasons,** but state mental-illness rates are not a guide to where mass shootings happen.

The takeaway

Where poverty and disadvantage concentrate, mass shootings concentrate too. Prevention resources belong in those places, and economic investment there is worth testing, alongside gun policy.

State data cannot settle whether gun laws work. State mental-illness rates are not a guide to where shootings happen. These are associations, not proof of cause.

Open data. The dataset, its sources, and the code that produces every number are free to use: Open Science Framework, doi.org/10.17605/OSF.IO/V5Y7E.

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